

Additional Models Explored: Synthetic Tabular Data Generation (Papers & GitHub)

1. TabPFGGen

- **Model Type:** Energy-Based Model (EBM)
- **Model Overview:**

TabPFGGen is a synthetic tabular data generation approach that leverages a pretrained TabPFN model within an energy-based modelling framework. The method generates synthetic samples by learning an implicit energy function over tabular data, enabling high-quality data generation without training a standalone generative network from scratch.
- **Research Paper:**
<https://arxiv.org/pdf/2406.05216>
- **GitHub Availability:**

The research paper does **not** provide an official GitHub repository for TabPFGGen.
- **Implementation Followed:**
<https://github.com/sebhaan/TabPFGGen>

2. Masked Ensemble Generator (MEG)

- **Model Type:** Ensemble-Based Generative Model
- **Model Overview:**

Masked Ensemble Generator (MEG) is a generative approach for tabular data that learns data distributions through masked reconstruction using an ensemble of models. The method operates by randomly masking subsets of features and training multiple learners to reconstruct the missing values, enabling robust modelling of complex feature dependencies and uncertainty through ensemble diversity.
- **Research Paper:**
https://www.nayyarzaidi.com/papers/MEG_Pre-publication_Draft.pdf
- **GitHub Availability:**

At the time of implementation, the research paper does **not** provide an official or publicly maintained GitHub repository for a tabular MEG implementation. Existing open-source repositories referencing “MEG” primarily target unrelated domains, such as image-based energy models or magnetoencephalography, and are not applicable to tabular synthetic data generation.
- **Implementation Followed:**

Due to the absence of an official implementation, MEG was implemented from scratch in this project based directly on the methodology described in the research paper. The implementation was adapted to the Katabatic benchmarking framework and follows the standard Katabatic model interface to ensure fair, consistent, and reproducible evaluation alongside other synthetic tabular data generators.

3. Masked Autoregressive Flow (MAF)

- **Model Type:** Normalising Flow (Autoregressive Flow)

- **Model Overview:**
Masked Autoregressive Flow (MAF) is a likelihood-based generative model that represents complex data distributions using a sequence of invertible autoregressive transformations. By stacking masked autoregressive layers, MAF enables exact density estimation with tractable Jacobians while maintaining high modelling flexibility. The model is particularly well-suited for density estimation tasks and forms a core class of normalising flow-based generative methods.
- **Research Paper:**
Papamakarios et al. (2017), *Masked Autoregressive Flow for Density Estimation*
<https://arxiv.org/abs/1705.07057>
- **GitHub Availability (Paper):**
The original research paper provides an experimental GitHub repository for reproducing the paper's results:
<https://github.com/gpapamak/maf>
- NIPS-2017-masked-autoregressive...
However, this implementation is Theano-based, unmaintained, and not suitable for modern PyTorch-based benchmarking pipelines.
- **Implementation Followed:**
In this project, MAF was implemented using the **nflows** library, which provides a research-grade, actively maintained PyTorch implementation of normalising flow models, including MAF. The implementation faithfully follows the original formulation proposed by Papamakarios et al. (2017) and was integrated into the Katabatic framework through a custom adapter to ensure consistent training, synthetic data generation, and fair evaluation under the TSTR protocol.
- **Implementation Repository:**
<https://github.com/bayesiains/nflows>

4. Adversarial Random Forests (ARF)

- **Model Type:** Forest-Based Generative Model
- **Model Overview:**
Adversarial Random Forests (ARF) is a non-neural, tree-based generative modelling approach that combines Random Forests with an adversarial training mechanism. The model alternates between generating synthetic samples and training a discriminator forest to distinguish real from synthetic data. Through this adversarial process, ARF learns an implicit data distribution that can be used for density estimation and synthetic tabular data generation without relying on deep neural networks.
- **Research Paper (Methodology):**
Watson et al. (2022), *Adversarial Random Forests*
<https://arxiv.org/abs/2205.09435>
- **Research Paper (Software / Implementation):**
Blesch, K. & Wright, M. N. (2023), *arfp: A Python Package for Density Estimation and Generative Modeling with Adversarial Random Forests*
<https://arxiv.org/abs/2311.07366>
- **GitHub Availability (Paper):**
The software paper explicitly provides an official GitHub repository for the ARF implementation.
- **Implementation Followed:**
The ARF model was implemented using **arfp**, the official Python package developed and maintained by the original authors. This implementation directly corresponds to the methodology described in the ARF research papers and provides a stable, research-grade

framework for density estimation and synthetic data generation. The model was integrated into the Katabatic benchmarking framework using a custom adapter to ensure consistent training, data generation, and fair TSTR-based evaluation.

- **Implementation Repository:**
<https://github.com/bips-hb/arfpv>

5. ForestDiffusion

- **Model Type:** Tree-Based Diffusion Model
- **Model Overview:**
ForestDiffusion is a diffusion-based generative model designed specifically for tabular data, where the denoising process is parameterised using tree-based learners instead of neural networks. The model replaces traditional neural diffusion backbones with ensembles of decision trees, enabling effective modelling of complex feature interactions while maintaining strong performance on structured tabular datasets. This design allows ForestDiffusion to leverage the strengths of tree-based methods within a diffusion framework for high-quality synthetic data generation.
- **Research Paper:**
ForestDiffusion: Tree-Based Diffusion Models for Tabular Data
<https://arxiv.org/abs/2309.09968>
- **GitHub Availability (Paper):**
The research paper explicitly provides an official GitHub repository for the ForestDiffusion implementation.
- **Implementation Followed:**
ForestDiffusion was implemented using the official GitHub repository provided by the authors. This implementation directly corresponds to the methodology described in the research paper and is maintained by the original authors. The model was integrated into the Katabatic benchmarking framework through a custom adapter, ensuring consistent training, synthetic data generation, and fair evaluation under the TSTR protocol.
- **Implementation Repository:**
<https://github.com/SamsungSAILMontreal/ForestDiffusion>

6. CopulaGAN

- **Model Type:** Hybrid (Statistical Copula + Generative Adversarial Network)
- **Model Overview:**
CopulaGAN is a hybrid synthetic tabular data generation model that integrates copula-based statistical dependency modelling with generative adversarial networks. The model separates the learning of marginal feature distributions from the modelling of inter-feature dependencies, using copulas to capture statistical relationships while a GAN refines complex, non-linear interactions. This design enables CopulaGAN to effectively generate realistic synthetic data for mixed-type tabular datasets while preserving both marginal properties and dependency structures.
- **Research Paper:**
CopulaGAN: Synthetic Tabular Data Generation via Copula and GAN
<https://arxiv.org/abs/2207.14406>
- **GitHub Availability (Paper):**
The research paper does **not** provide an official GitHub repository for CopulaGAN.

- **Implementation Followed:**
CopulaGAN was implemented using the official **SDV (Synthetic Data Vault)** library, which provides a maintained and research-aligned implementation of the model. The SDV implementation represents the authoritative reference for CopulaGAN and is widely used in both academic research and practical applications. In this project, CopulaGAN was executed using default SDV configurations without hyperparameter tuning, ensuring fidelity to the published methodology and consistency across all benchmarking experiments within the Katabatic framework.
- **Implementation Repository:**
<https://github.com/sdv-dev/SDV>

7. TabKDE

- **Model Type:** Statistical Density-Based Model (KDE with Copula Transformations)
- **Model Overview:**
TabKDE is a simple and scalable synthetic tabular data generation method based on kernel density estimation (KDE) applied in a copula-transformed latent space. The model first maps tabular features into a latent representation using copula transformations, enabling effective handling of mixed data types. Synthetic samples are then generated via KDE-based resampling, optionally combined with Gaussian mixture modelling. TabKDE is a non-neural, non-iterative approach that requires no gradient-based optimisation or epoch-based training, making it a fast and interpretable baseline for tabular data synthesis.
- **Research Paper:**
Elor, Y., Avineri, G., & Lederman, R. (2023), *TabKDE: Simple and Scalable KDE-based Tabular Data Generation*
<https://arxiv.org/abs/2305.11527>
- **GitHub Availability (Paper):**
The research paper explicitly provides an official GitHub repository for the TabKDE implementation.
- **Implementation Followed:**
TabKDE was implemented following the original methodology described by Elor et al. (2023) using the official GitHub repository provided by the authors. The implementation adheres strictly to the paper's design principles, including single-pass fitting, unconditional generation, and the absence of hyperparameter tuning or dataset-specific optimisation. A thin adapter layer was introduced solely to integrate TabKDE into the Katabatic benchmarking framework, enabling consistent data splitting and evaluation under the TSTR protocol without modifying the underlying algorithm.
- **Implementation Repository:**
<https://github.com/tabkde/tabkde-main>

8. PrivBayes

- **Model Type:** Differentially Private Bayesian Network Model
- **Model Overview:**
PrivBayes is a differentially private synthetic tabular data generation method based on Bayesian networks. The model learns a probabilistic graphical structure over the data using the exponential mechanism to ensure differential privacy during structure

selection. Laplace noise is added to conditional probability tables, and synthetic records are generated by sampling from the noisy Bayesian network. PrivBayes is a non-neural, non-iterative approach and does not rely on epoch-based training.

- **Research Paper:**
Zhang, J., Cormode, G., Procopiuc, C., Srivastava, D., & Xiao, X. (2014), *PrivBayes: Private Data Release via Bayesian Networks*
<https://dl.acm.org/doi/10.1145/2588555.2588573>
- **GitHub Availability (Paper):**
The research paper does **not** provide an official GitHub repository for PrivBayes.
- **Implementation Followed:**
PrivBayes was implemented as a clean reimplement based directly on the original ACM SIGMOD 2014 paper. The implementation follows all algorithmic decisions described in the paper, including greedy Bayesian network construction, exponential mechanism-based parent selection, and Laplace noise injection controlled by the privacy parameter ϵ . The widely cited **DataSynthesizer** project was used only as a reference implementation for validation purposes, as no official code release exists. The model was integrated into the Katabatic framework via a standard adapter to support fair and reproducible TSTR-based evaluation.
- **Reference Implementation (Validation Only):**
<https://github.com/DataResponsibly/DataSynthesizer>

9. Quantile Regression Forests (QRF)

- **Model Type:** Tree-Based Distributional Regression Model
- **Model Overview:**
Quantile Regression Forests (QRF) is a non-parametric, tree-ensemble-based method for estimating conditional distributions rather than point predictions. Extending the Random Forest framework, QRF retains the full distribution of target values within tree leaves, enabling the estimation of arbitrary conditional quantiles. This allows QRF to model uncertainty and distributional properties of tabular data without relying on parametric assumptions, neural networks, or iterative optimisation.
- **Research Paper:**
Meinshausen, N. (2006), *Quantile Regression Forests*
<https://www.jmlr.org/papers/volume7/meinshausen06a/meinshausen06a.pdf>
- **GitHub Availability (Paper):**
The original research paper does **not** provide an official GitHub repository for Quantile Regression Forests.
- **Implementation Followed:**
As no official Python implementation was released by the original author, QRF was implemented using the widely adopted **quantile-forest** Python library maintained by Zillow. This implementation is scikit-learn compatible and faithfully follows the original formulation proposed by Meinshausen (2006). The model was integrated into the Katabatic framework via a lightweight adapter to enable consistent training, sampling, and fair TSTR-based benchmarking without modifying the underlying algorithm or tuning hyperparameters.
- **Implementation Repository:**
<https://github.com/zillow/quantile-forest>

10. CTAB-GAN+

- **Model Type:** Conditional Generative Adversarial Network (GAN) for Tabular Data
- **Model Overview:**
CTAB-GAN+ is an advanced conditional GAN framework designed for high-quality tabular data generation. The model extends the original CTAB-GAN by improving the handling of mixed data types, rare categorical values, and imbalanced classes. It employs a conditional vector mechanism to guide data generation, along with specialised encoders for numerical and categorical features. CTAB-GAN+ uses a Wasserstein GAN with gradient penalty (WGAN-GP) to stabilise training and improve convergence, making it suitable for complex real-world tabular datasets.
- **Research Paper:**
Xu, L. et al. (2021), *CTAB-GAN+: Improving Tabular Data Generation Using Conditional GANs*
<https://arxiv.org/abs/2204.00401>
- **GitHub Availability (Paper):**
The research paper explicitly provides an official GitHub repository for CTAB-GAN+.
- **Implementation Followed:**
CTAB-GAN+ was implemented using the official GitHub repository released by the original authors. The implementation strictly follows the architecture, training strategy, and default hyperparameters specified in the paper (including conditional vectors and training configuration). No hyperparameter tuning or architectural modifications were applied. The model was integrated into the Katabatic benchmarking framework via a standard adapter to enable consistent training, synthetic data generation, and fair evaluation using the TSTR protocol.
- **Implementation Repository:**
<https://github.com/Team-TUD/CTAB-GAN-Plus>

11. SynthPop

- **Model Type:** Statistical Sequential Data Synthesis Model
- **Model Overview:**
SynthPop is a non-neural, statistical synthetic tabular data generation method based on sequential conditional modelling. The model synthesises data one variable at a time by fitting conditional models (such as CART and GLM) to observed data and sampling from the resulting distributions. SynthPop operates in a single-pass manner without epochs or gradient-based optimisation, making it a fast and interpretable baseline for tabular data synthesis.
- **Research Paper:**
Nowok, B., Raab, G. M., & Dibben, C. (2016), *Synthpop: Bespoke Creation of Synthetic Data in R*
<https://www.jstatsoft.org/article/view/v074i11>
- **GitHub Availability (Paper):**
The research paper documents the SynthPop R package, which is maintained by the original authors.
- **Implementation Followed:**
SynthPop was implemented using the official GitHub repository released and

maintained by the original authors. The implementation follows the methodology described in the Journal of Statistical Software paper exactly, with no hyperparameter tuning or algorithmic modification. A lightweight adapter was used solely to integrate SynthPop into the Katabatic benchmarking framework and enable consistent evaluation under the TSTR protocol.

- **Implementation Repository:**
<https://github.com/bnowok/synthpop>

12.DECAF

- **Model Type:** Causally-Aware Generative Adversarial Network (GAN)
- **Model Overview:**
DECAF is a causally-aware generative model designed to produce fair synthetic tabular data by explicitly incorporating causal relationships between variables. The model combines structural causal models with generative adversarial networks to ensure that sensitive attributes and downstream variables follow causal constraints rather than purely statistical correlations. DECAF learns a directed acyclic graph (DAG) representing causal dependencies and generates synthetic data consistent with this structure, enabling fairness-aware data synthesis without post-hoc corrections.
- **Research Paper:**
van Breugel, B., Kyono, T., Berrevoets, J., & van der Schaar, M. (2021), *DECAF: Generating Fair Synthetic Data Using Causally-Aware Generative Networks*
NeurIPS 2021
<https://arxiv.org/abs/2110.12884>
- **GitHub Availability (Paper):**
The research paper provides an official GitHub repository for DECAF.
- **Implementation Followed:**
DECAF was implemented using the official GitHub repository released by the original authors and maintained by the van der Schaar Lab. The implementation corresponds directly to the code used in the NeurIPS 2021 paper and follows the paper-specified training configuration (including 300 training epochs) without any hyperparameter tuning or architectural modifications. The model was integrated into the Katabatic benchmarking framework via a standard adapter to support consistent synthetic data generation and fair evaluation under the TSTR protocol.
- **Implementation Repository:**
<https://github.com/vanderschaarlab/DECAF>

13.GOGGLE

- **Model Type:** Graph-Based Generative Model for Tabular Data
- **Model Overview:**
GOGGLE (Graph-Oriented Generative Model for Tabular Data) is a graph-based generative framework designed to synthesise tabular data by explicitly modelling dependencies between features as a learned graph structure. The model represents tabular attributes as nodes in a graph and learns their relationships through a graph encoder-decoder architecture. By combining graph neural networks with generative

modelling, GOGGLE captures complex inter-feature dependencies beyond pairwise correlations, enabling high-fidelity synthetic data generation for structured datasets.

- **Research Paper:**
Liu, W., Huang, T.-H., & van der Schaar, M. (2023),
GOGGLE: Graph-Oriented Generative mOdeL for Tabular Data
<https://arxiv.org/abs/2306.13003>
- **GitHub Availability (Paper):**
The research paper explicitly provides an official GitHub repository for the GOGGLE implementation.
- **Implementation Followed:**
GOGGLE was implemented using the official GitHub repository released and maintained by the original authors from the van der Schaar Lab. The implementation corresponds directly to the architecture and optimisation strategy described in the paper, including the use of DGL and PyTorch for graph learning and generation. All experiments were conducted using paper-default settings, with training performed for 1000 epochs and no hyperparameter tuning or architectural modifications applied. The model was integrated into the Katabatic benchmarking framework via a standard adapter to ensure consistent synthetic data generation and fair evaluation under the TSTR protocol.
- **Implementation Repository:**
<https://github.com/tennisonliu/GOGGLE>

14. TAEGAN

- **Model Type:** Autoencoder-Based Generative Adversarial Network for Tabular Data
- **Model Overview:**
TAEGAN is a generative adversarial network designed for tabular data synthesis that integrates an autoencoder with a GAN framework. The model first learns a compact latent representation of tabular data using an autoencoder, which helps stabilise training and preserve feature semantics. A GAN is then trained in the latent space to generate realistic synthetic samples, which are decoded back into the original feature space. This architecture improves data fidelity and training stability, particularly for mixed-type tabular datasets.
- **Research Paper:**
Li, J. et al., *TAEGAN: Tabular Data Generation Using Autoencoder-based GAN*
<https://openreview.net/forum?id=bWUul8rjqZ>
- **GitHub Availability (Paper):**
The research paper provides an official GitHub repository for the TAEGAN implementation.
- **Implementation Followed:**
TAEGAN was implemented using the official GitHub repository released and maintained by the original authors. The implementation corresponds directly to the architecture and training procedure described in the paper. All experiments were conducted using paper-default settings, with no hyperparameter tuning or architectural modifications applied. The model was integrated into the Katabatic benchmarking framework via a standard adapter to enable consistent synthetic data generation and fair evaluation under the TSTR protocol.
- **Implementation Repository:**
<https://github.com/BetterdataLabs/taegan>

- **NOTE:** All model implementations and corresponding code files have been fully completed and are ready for execution. However, full-scale execution was halted due to extreme computational cost, with training taking approximately 8-9 hours and causing memory (RAM) exhaustion even on small datasets. To prevent system instability and ensure project continuity, model execution was stopped. The implementations themselves are complete and can be executed in a higher-memory or dedicated compute environment.

15. TabEBM

- **Model Type:** Energy-Based Model (EBM) for Tabular Data
- **Model Overview:**
TabEBM is an energy-based generative model designed for tabular data synthesis. The model learns an unnormalised probability distribution by assigning lower energy values to realistic data samples and higher energy values to implausible ones. Unlike GANs or VAEs, TabEBM relies on iterative energy minimisation and sampling procedures, which are computationally intensive but allow flexible modelling of complex tabular distributions. Energy-based models are expressive but require substantial compute and memory resources during training and sampling.
- **Research Paper:**
TabEBM: Energy-Based Models for Tabular Data Generation
<https://arxiv.org/abs/2409.16118>
- **GitHub Availability (Paper):**
The research paper provides an official GitHub repository for the TabEBM implementation.
- **Implementation Repository:**
<https://github.com/andreimargeloiu/TabEBM>
- **Implementation Followed:**
TabEBM was integrated following the official research paper and the author-provided GitHub implementation. The model architecture, training logic, and sampling procedures were kept consistent with the paper, and no hyperparameter tuning or architectural modifications were applied. The implementation was wrapped in a Katabatic-compatible adapter to support standardised pipelines and evaluation logic.
- **NOTE:** All TabEBM model files and pipeline integrations have been fully completed and are ready. However, due to unresolved dependency and environment compatibility constraints, the model could not be executed on the benchmark datasets. Aside from execution, the model integration and pipeline setup are complete and can be run once the required dependencies are successfully configured in an appropriate compute environment.